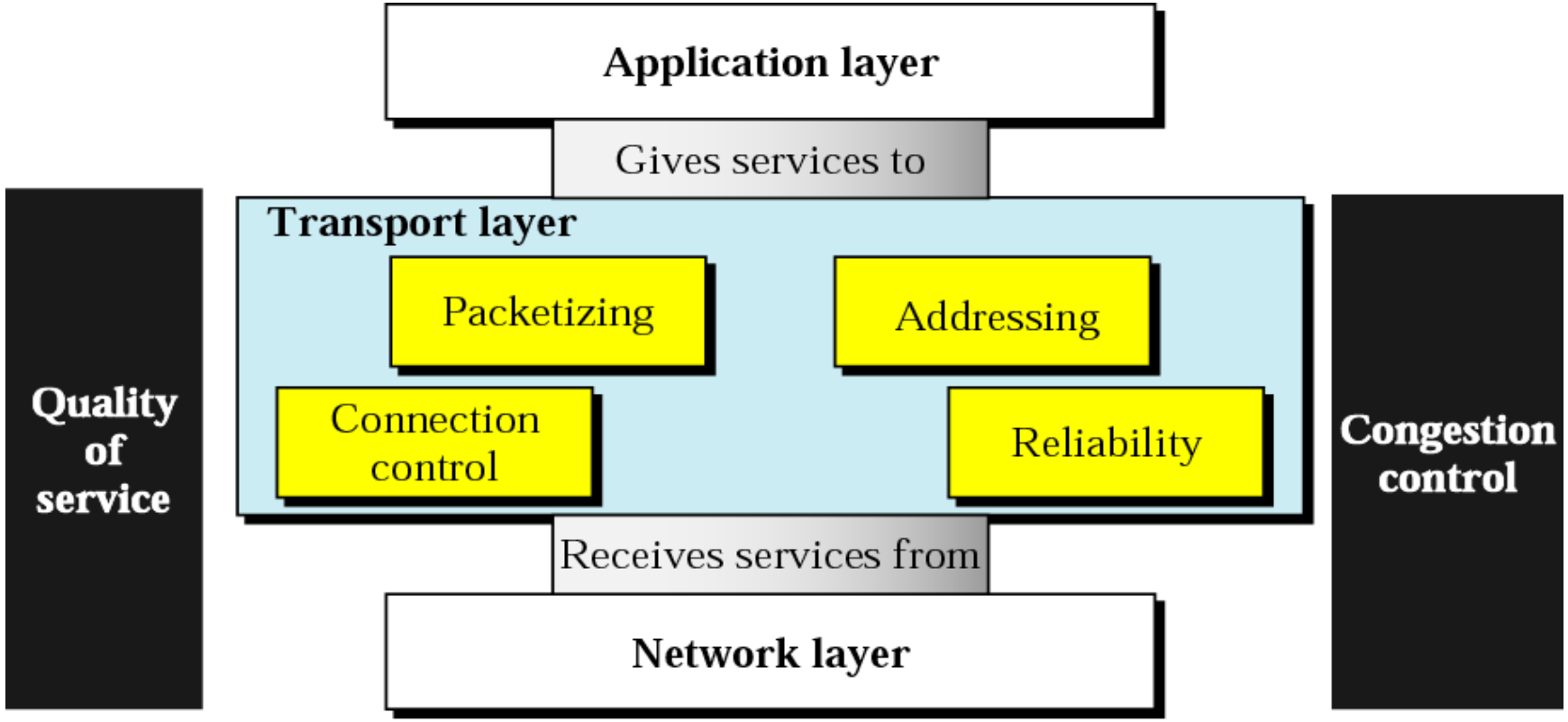
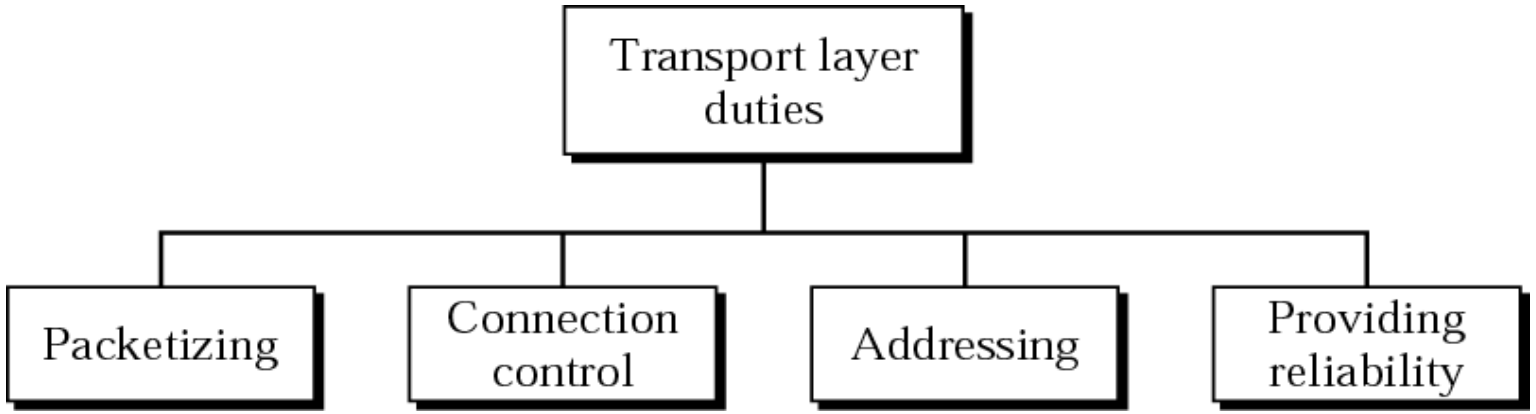
TRANSPORT LAYER



Position of transport layer



Transport layer duties



Process-to-Process Delivery:
UDP and TCP

Process-to-Process Delivery

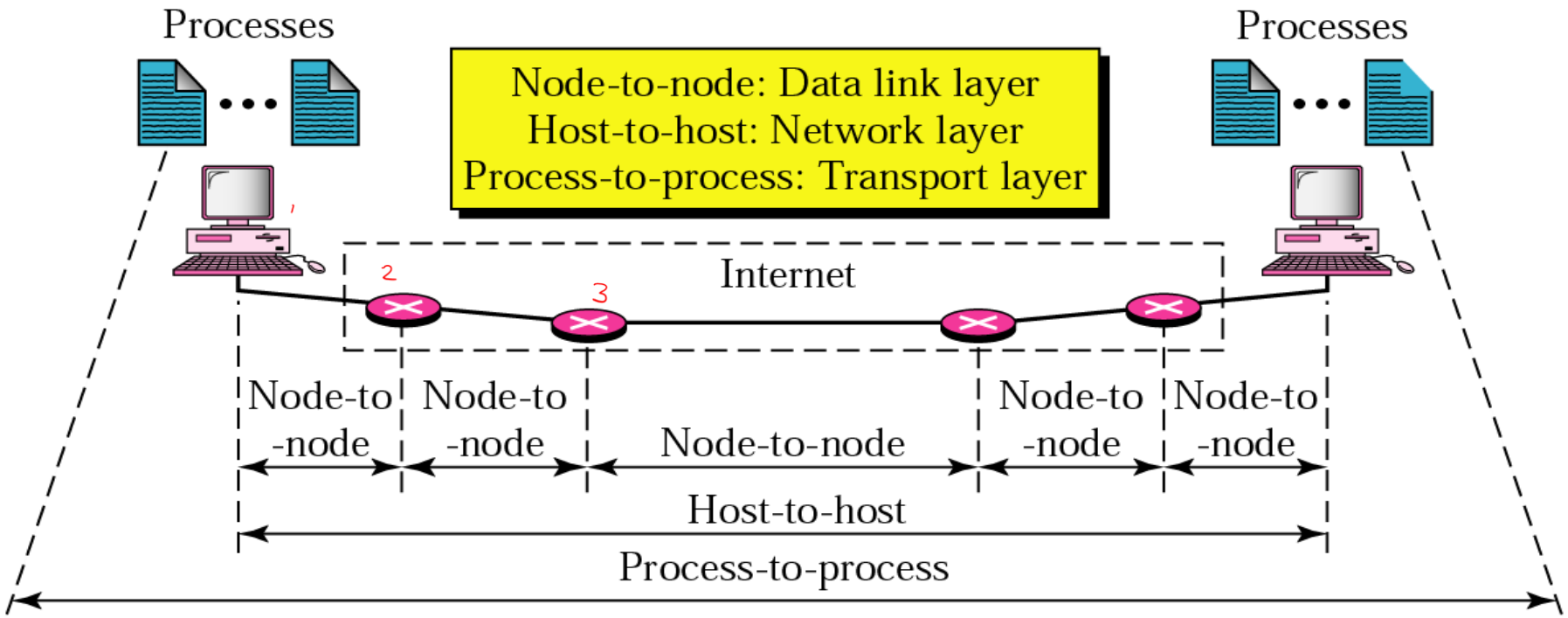
- Client-Server Paradigm
- Addressing
- Multiplexing and Demultiplexing
- Connectionless/Connection-Oriented
- Reliable/Unreliable



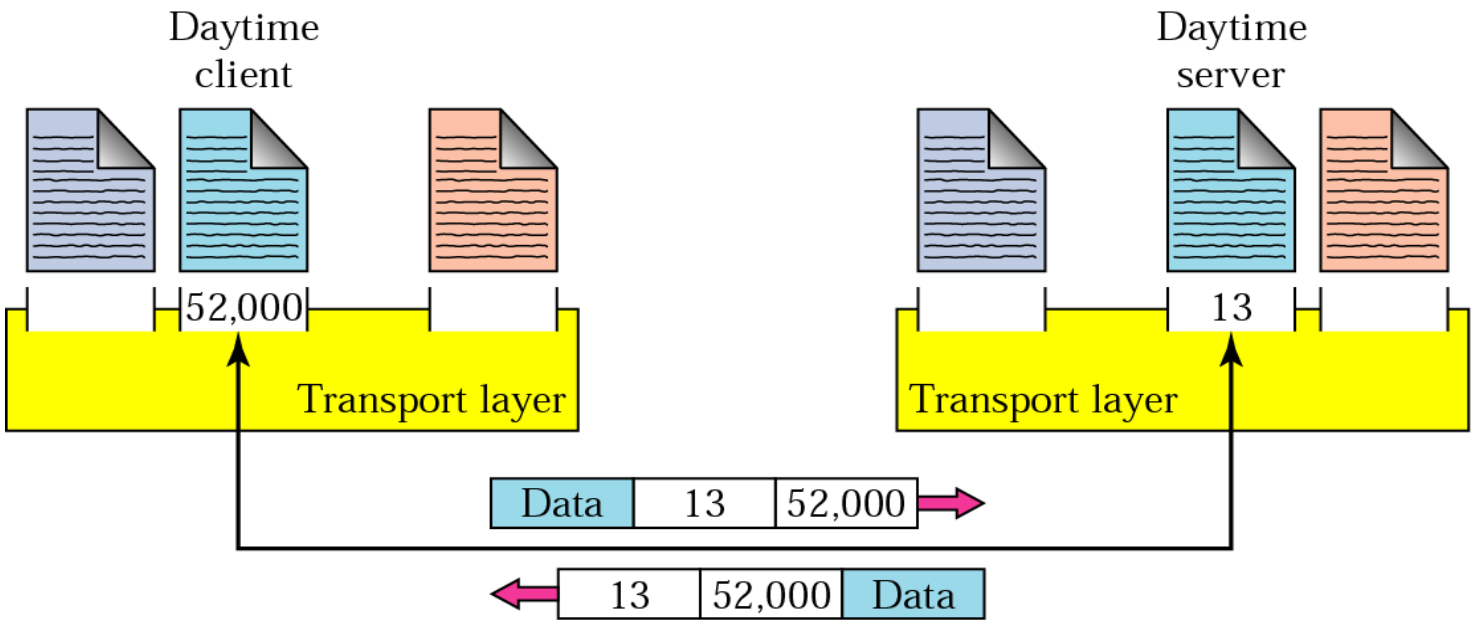
Note:

The transport layer is responsible for process-to-process delivery.

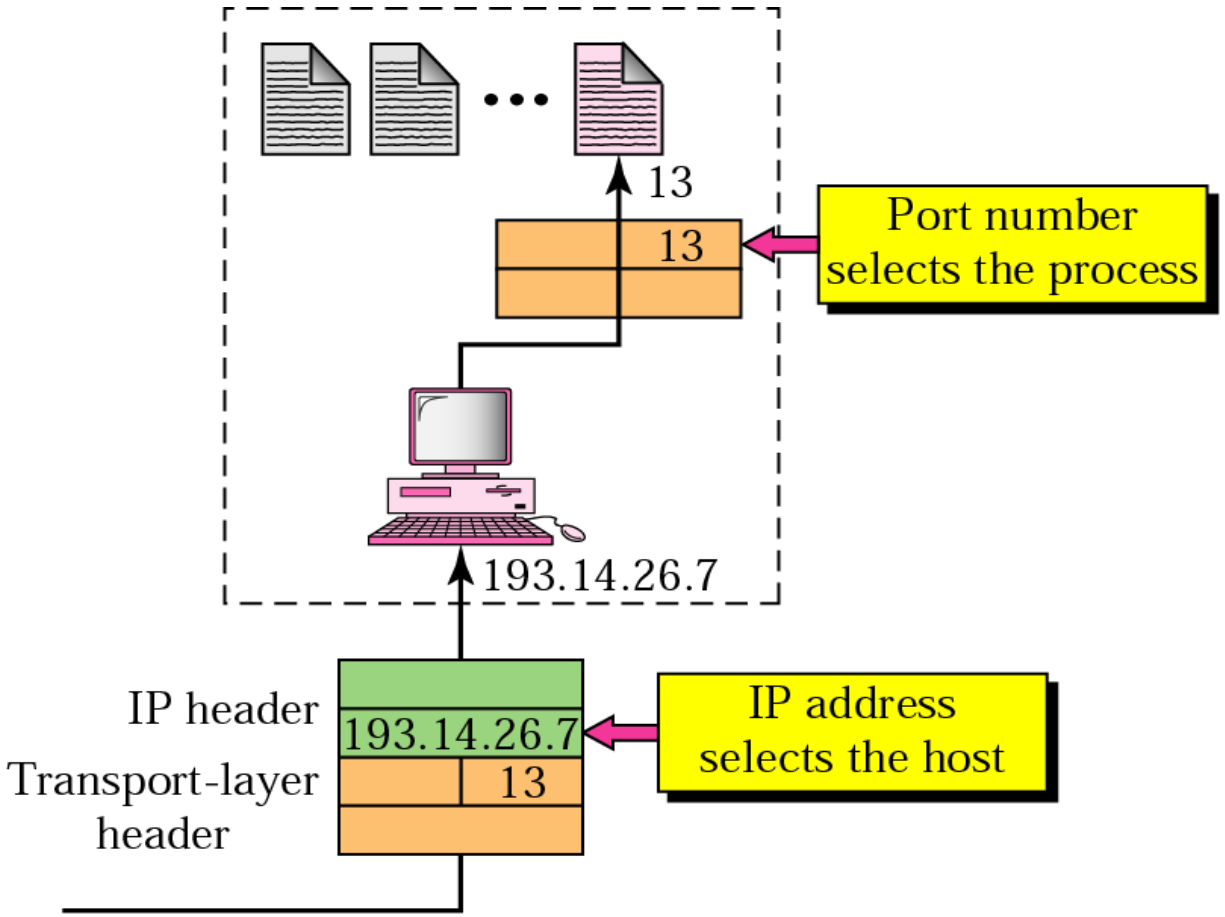
Types of data deliveries



Port numbers

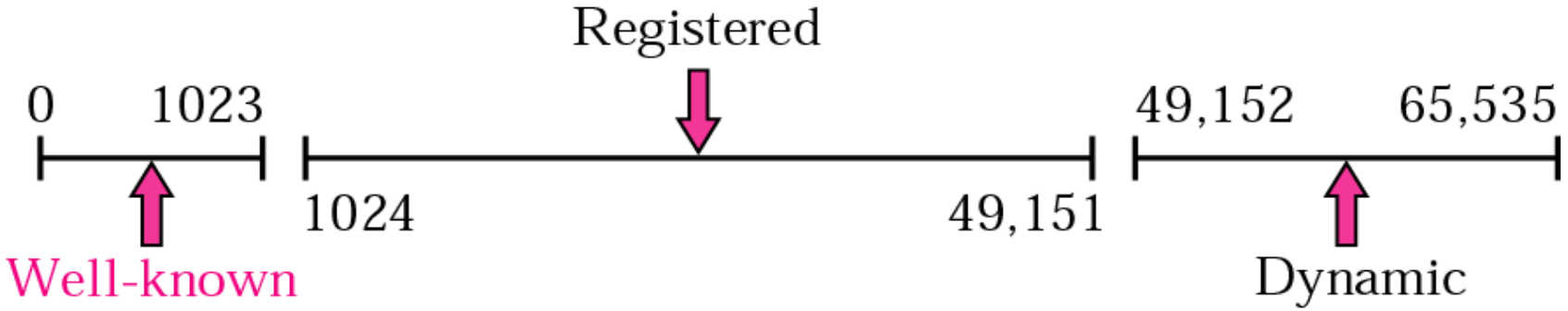


IP addresses versus port numbers

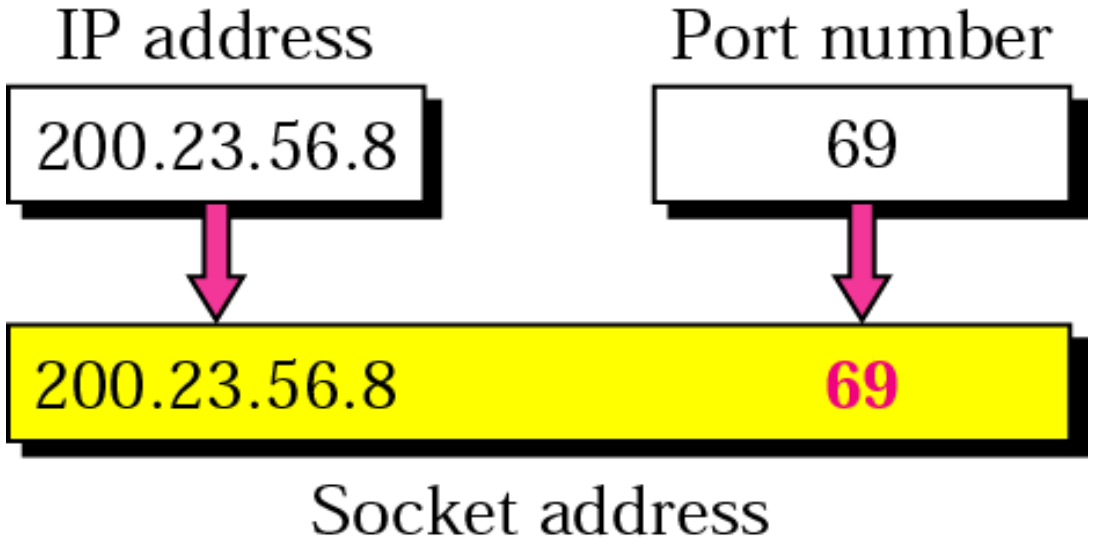


IANA ranges

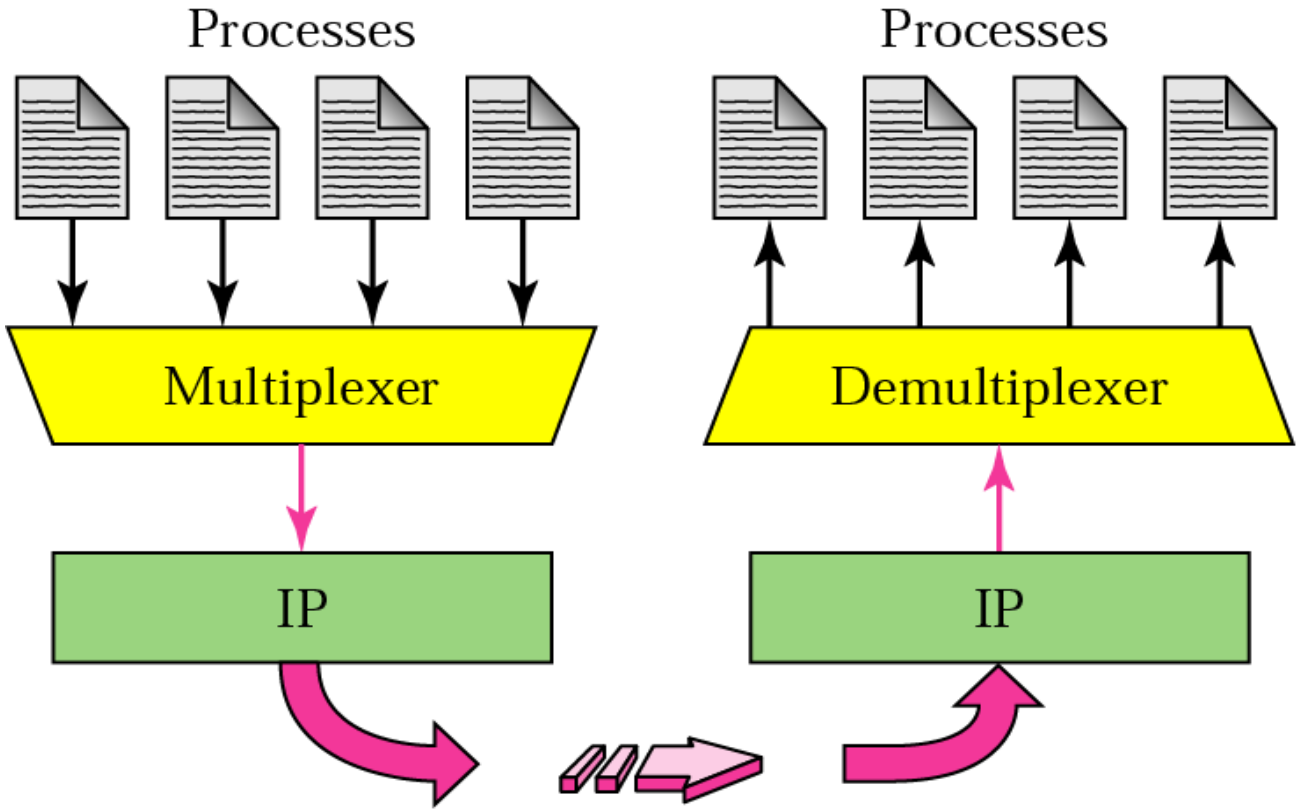
The IANA (Internet Assigned Number Authority) has divided the port numbers into three ranges: well known, registered, and dynamic (or private).



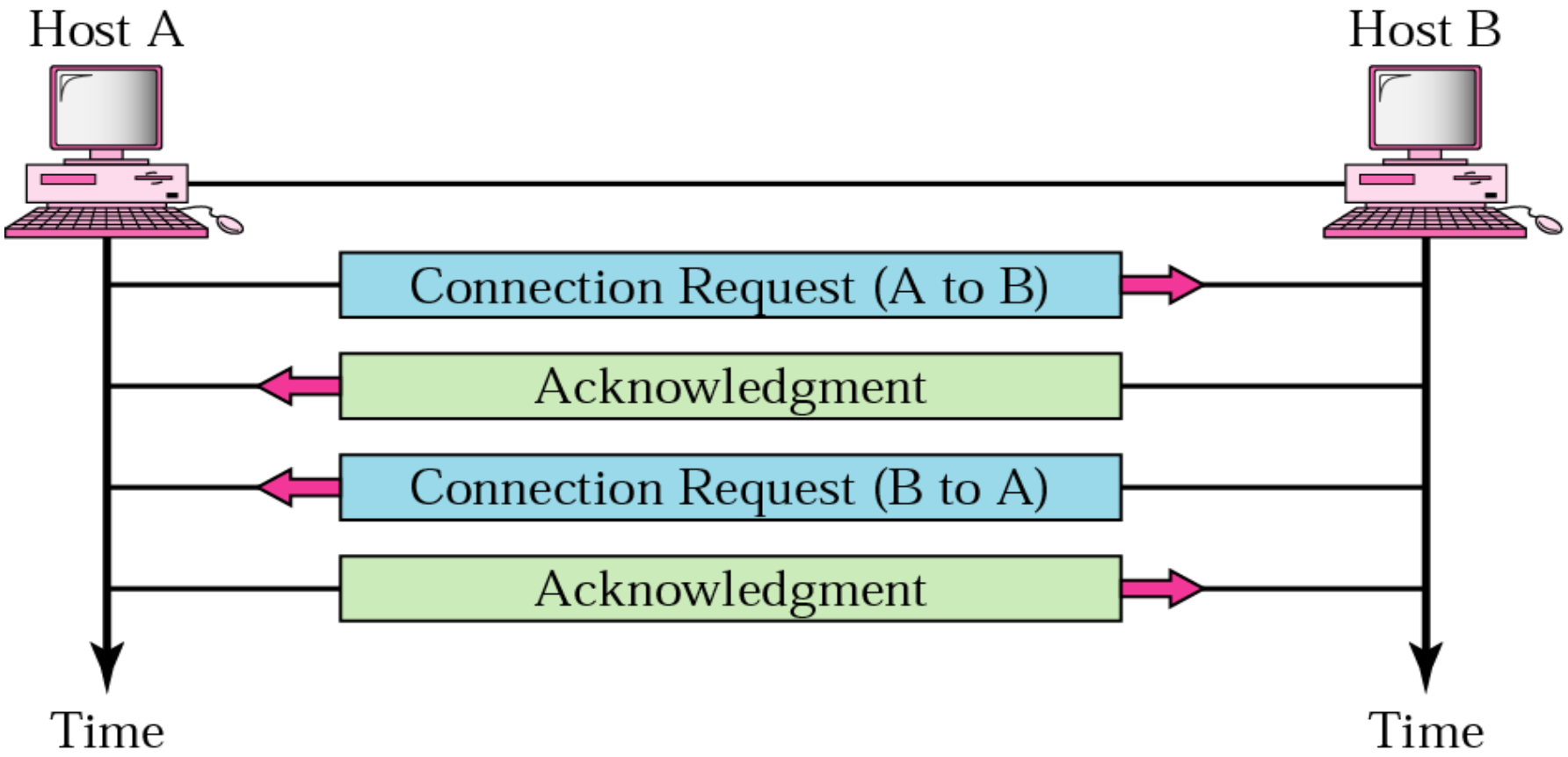
Socket address



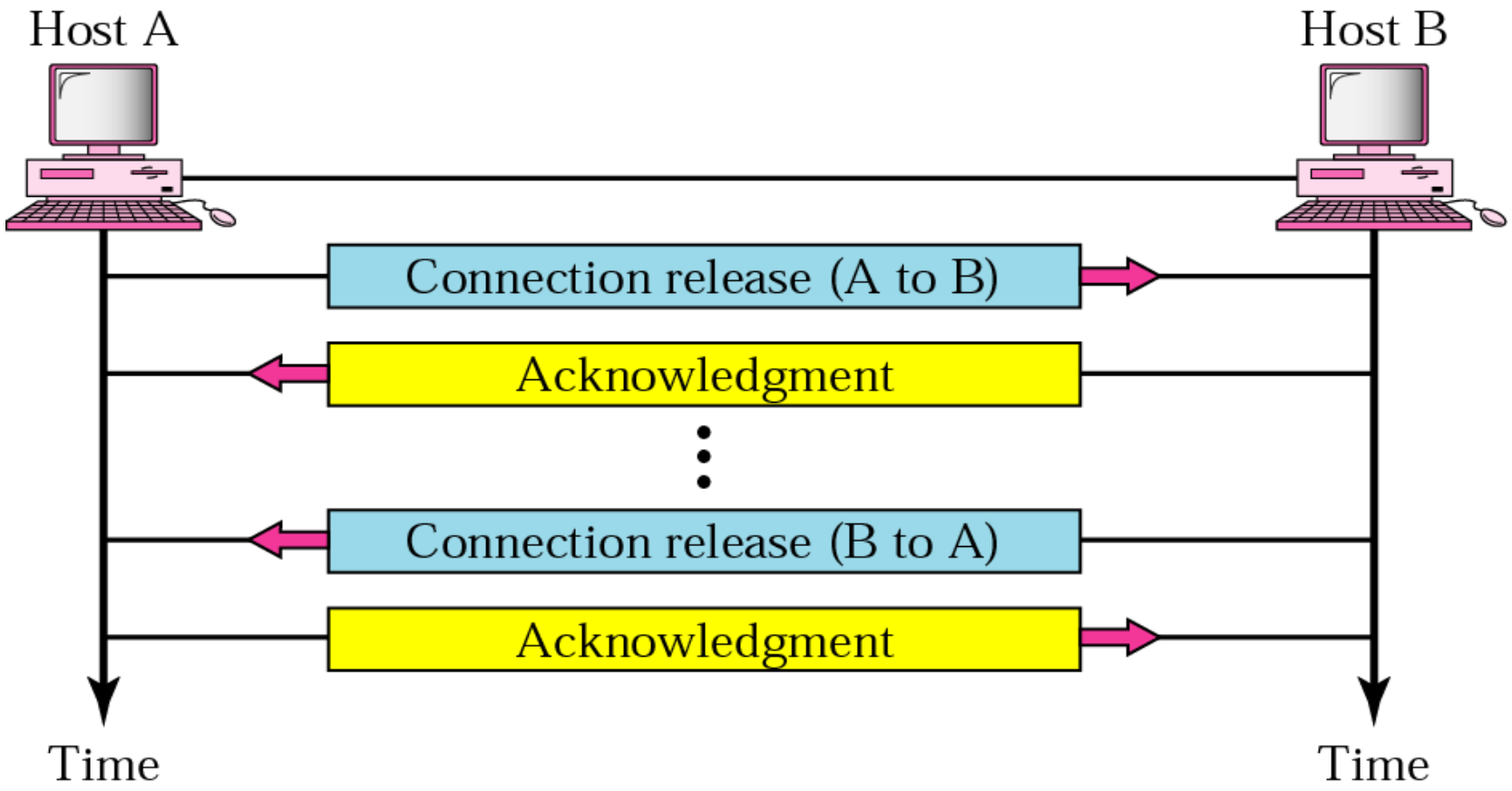
Multiplexing and demultiplexing



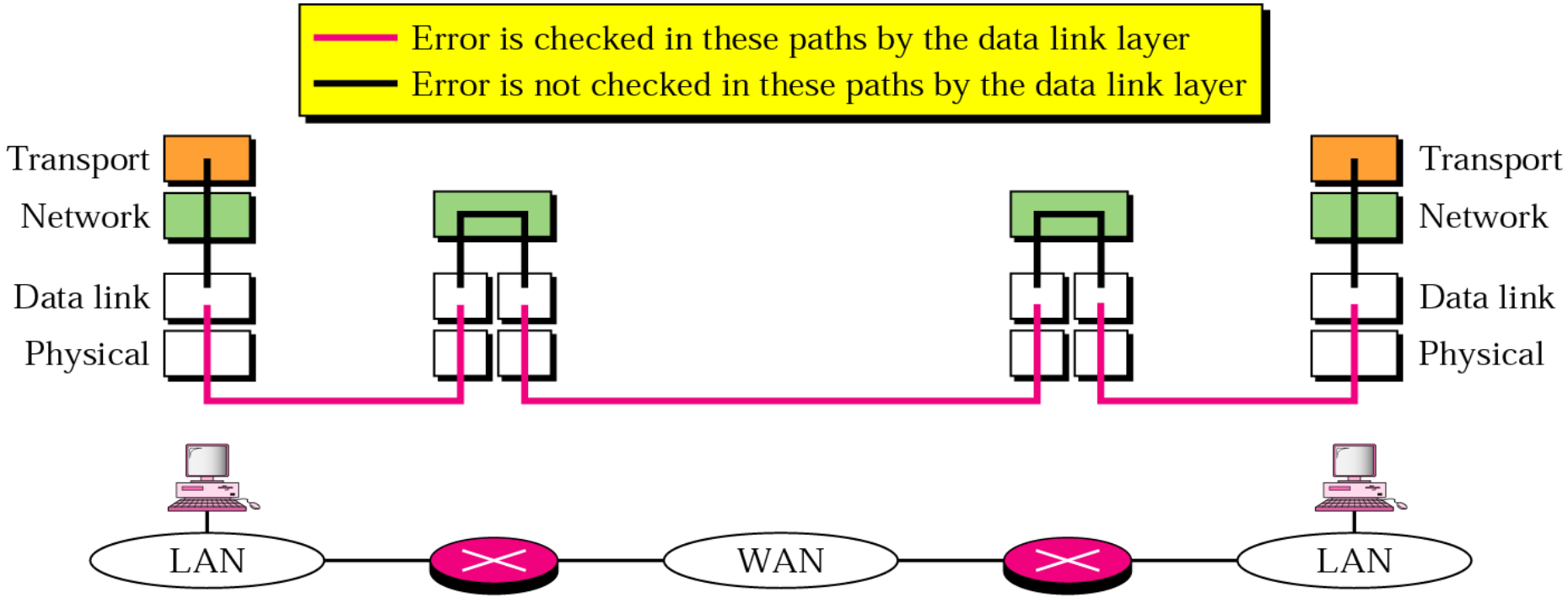
Connection establishment



Connection termination



Error control



UDP

- ***Port Numbers***
- ***User Datagram***
- ***Applications***

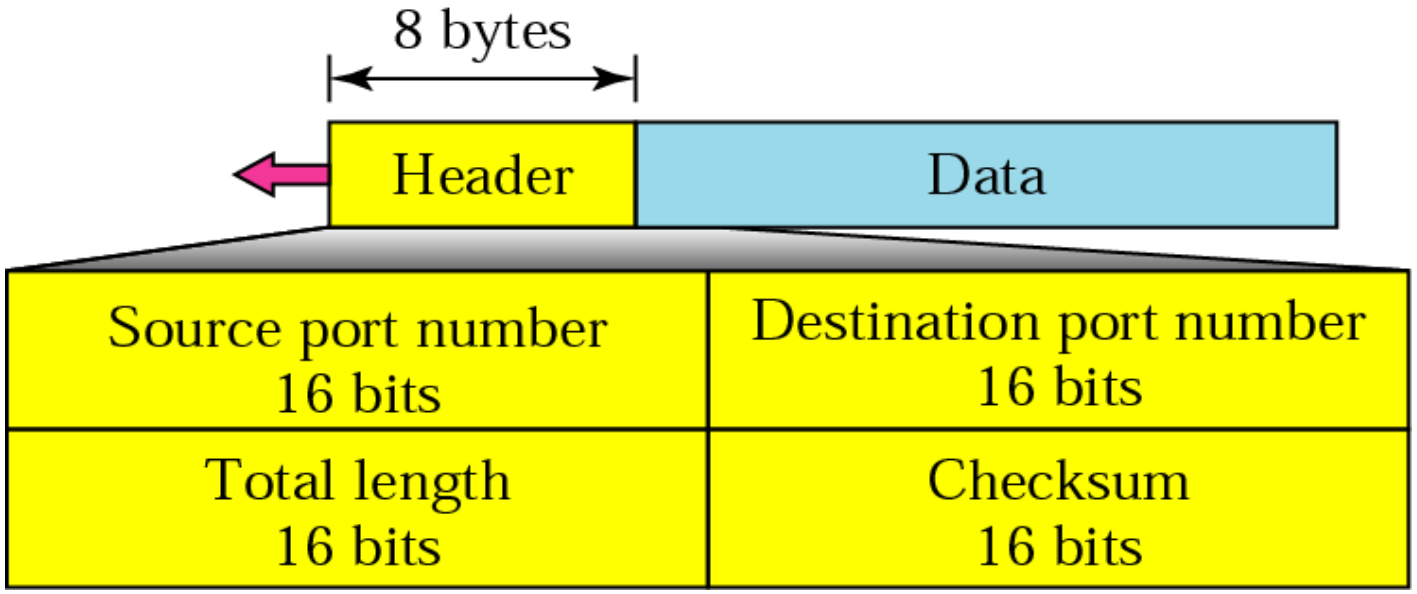


UDP is a connectionless, unreliable protocol that has no flow and error control. It uses port numbers to multiplex data from the application layer.

Well-known ports used by UDP

Port	Protocol	Description
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
53	Nameserver	Domain Name Service
67	Bootps	Server port to download bootstrap information
68	Bootpc	Client port to download bootstrap information
69	TFTP	Trivial File Transfer Protocol
111	RPC	Remote Procedure Call
123	NTP	Network Time Protocol
161	SNMP	Simple Network Management Protocol
162	SNMP	Simple Network Management Protocol (trap)

User datagram format





Note:

The calculation of checksum and its inclusion in the user datagram are optional.



Note:

UDP is a convenient transport-layer protocol for applications that provide flow and error control. It is also used by multimedia applications.

TCP – TRANSMISSION CONTROL PROTOCOL

INTRODUCTION

- Transmission Control Protocol
- TCP is a connection oriented protocol; it creates a virtual connection between two TCPs to send data.
- In addition, TCP uses flow and error control mechanisms at the transport level.
- TCP is called a connection-oriented, reliable transport protocol.
- It adds connection-oriented and reliability features to the services of IP.

TCP

- *Port Numbers*
- *Services*
- *Sequence Numbers*
- *Segments*
- *Connection*
- *Transition Diagram*
- *Flow and Error Control*
- *Silly Window Syndrome*

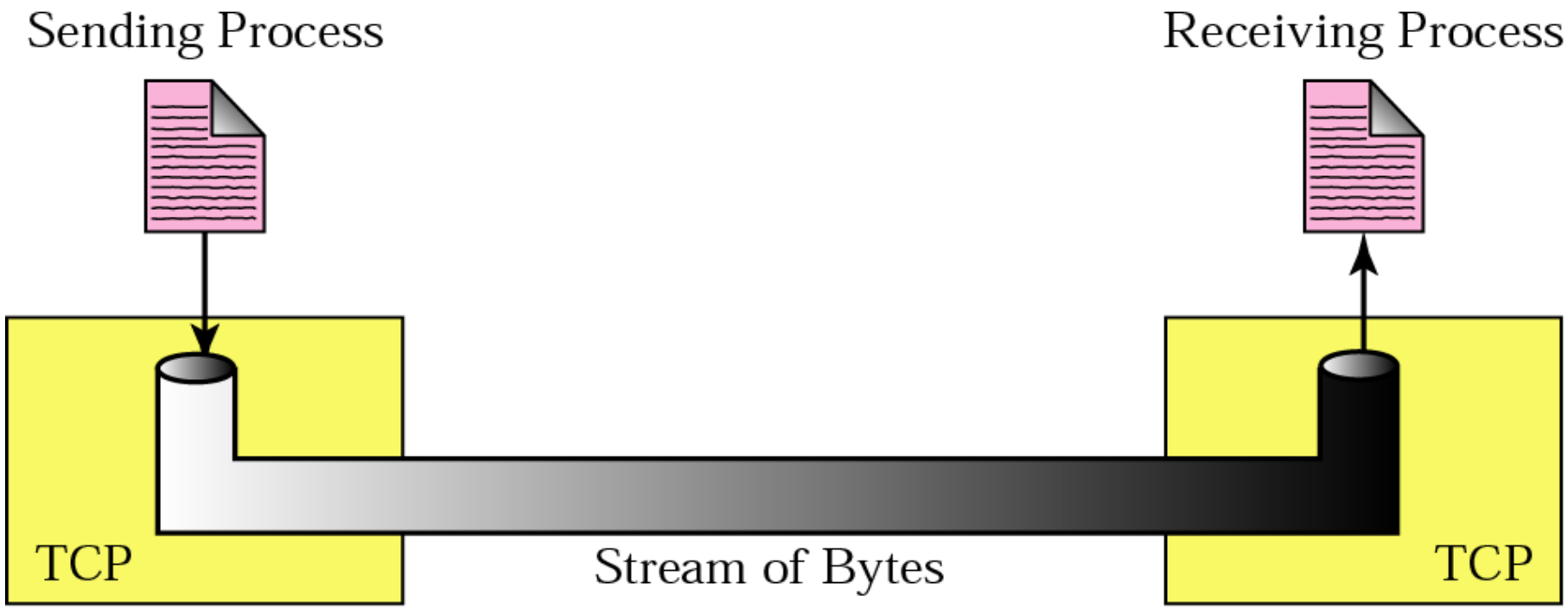
Well-known ports used by TCP

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11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
20	FTP, Data	File Transfer Protocol (data connection)
21	FTP, Control	File Transfer Protocol (control connection)
23	TELNET	Terminal Network
25	SMTP	Simple Mail Transfer Protocol
53	DNS	Domain Name Server
67	BOOTP	Bootstrap Protocol
79	Finger	Finger
80	HTTP	Hypertext Transfer Protocol
111	RPC	Remote Procedure Call

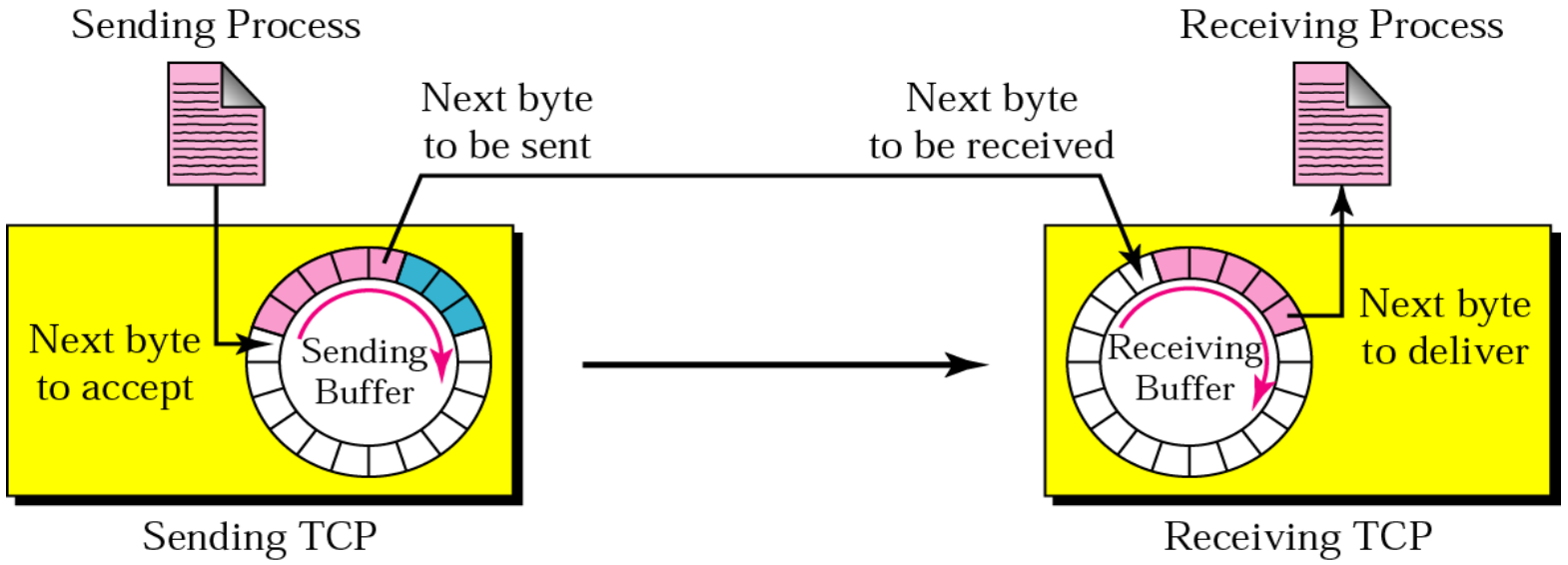
STREAM DELIVERY

- TCP allows the sending process to deliver data as a stream of bytes and allows the receiving process to obtain data as a stream of bytes.
- TCP creates an environment in which the two processes seem to be connected by an imaginary "tube" that carries their data across the Internet.
- The sending process produces (writes to) the stream of bytes, and the receiving process consumes (reads from) them.

Stream delivery



Sending and receiving buffers



CONTD…

- There are two buffers, the sending buffer and the receiving buffer, one for each direction.
- One way to implement a buffer is to use a circular array of I-byte locations.
- For simplicity, two buffers of 20 bytes each; normally the buffers are hundreds or thousands of bytes, depending on the implementation.
- Figure shows the movement of the data in one direction. At the sending site, the buffer has three types of chambers.
- The white section contains empty chambers that can be filled by the sending process (producer).

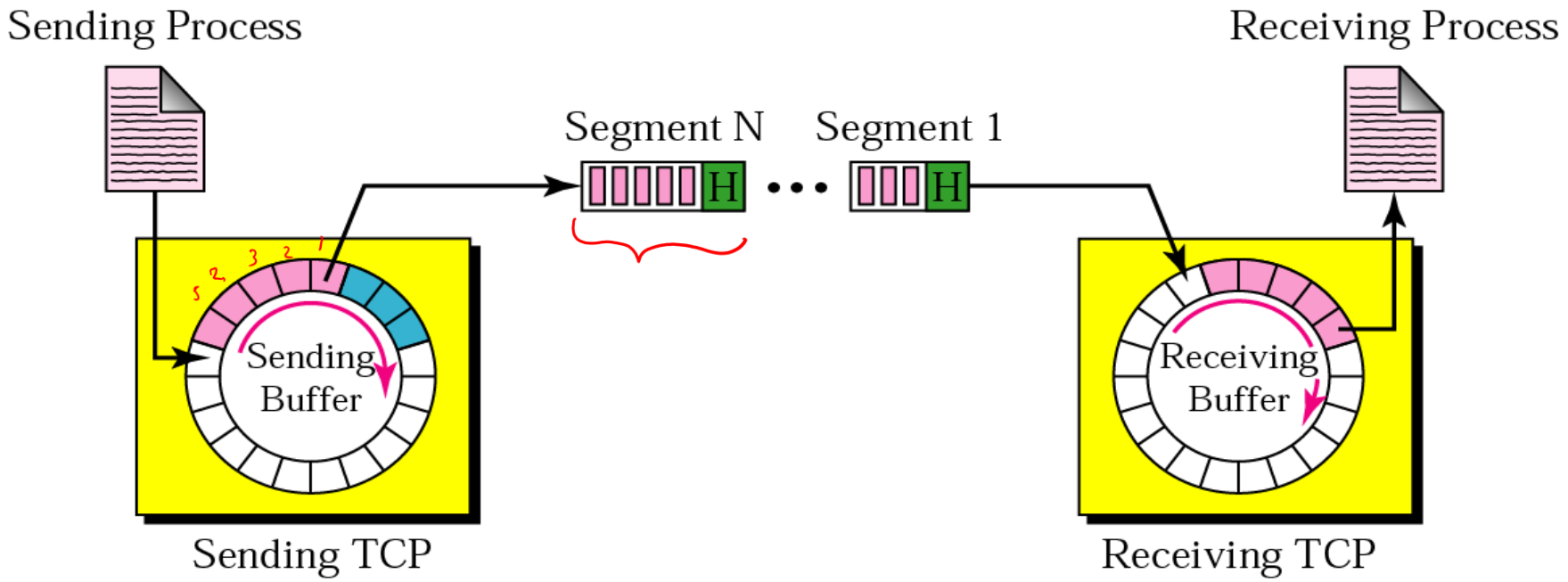
CONTD…

- The gray area holds bytes that have been sent but not yet acknowledged.
- TCP keeps these bytes in the buffer until it receives an acknowledgment.
- The colored area contains bytes to be sent by the sending TCP
- The operation of the buffer at the receiver site is simpler.
- The circular buffer is divided into two areas (shown as white and colored).

CONTD…

- The white area contains empty chambers to be filled by bytes received from the network.
- The colored sections contain received bytes that can be read by the receiving process.
- When a byte is read by the receiving process, the chamber is recycled and added to the pool of empty chambers.

TCP segments



CONTD…

- Although buffering handles the disparity between the speed of the producing and consuming processes, we need one more step before we can send data.
- The IP layer, as a service provider for TCP, needs to send data in packets, not as a stream of bytes.
- The transport layer, TCP groups a number of bytes together into a packet called a segment.
- TCP adds a header to each segment (for control purposes) and delivers the segment to the IP layer for transmission.
- The segments are encapsulated in IP datagrams and transmitted.
- This entire operation is transparent to the receiving process

Full-Duplex Communication

- TCP offers full-duplex service, in which data can flow in both directions at the same time.
- Each TCP then has a sending and receiving buffer, and segments move in both directions.

Connection-Oriented Service

- TCP, unlike UDP, is a connection-oriented protocol.
- When a process at site A wants to send and receive data from another process at site B, the following occurs:
    1. The two TCPs establish a connection between them.
    2. Data are exchanged in both directions.
    3. The connection is terminated.

CONTD…

- This is a virtual connection, not a physical connection.
- The TCP segment is encapsulated in an IP datagram and can be sent out of order, or lost, or corrupted, and then resent. Each may use a different path to reach the destination.
- There is no physical connection.
- TCP creates a stream-oriented environment in which it accepts the responsibility of delivering the bytes in order to the other site.
- The situation is similar to creating a bridge that spans multiple islands and passing all the bytes from one island to another in one single connection.

CONTD…

Reliable Service

- TCP is a reliable transport protocol.
- It uses an acknowledgment mechanism to check the safe and sound arrival of data.

Example 1

Imagine a TCP connection is transferring a file of 6000 bytes. The first byte is numbered 10010. What are the sequence numbers for each segment if data are sent in five segments with the first four segments carrying 1000 bytes and the last segment carrying 2000 bytes?

Solution

The following shows the sequence number for each segment:

Segment 1 ==> sequence number: 10,010 (range: 10,010 to 11,009)
Segment 2 ==> sequence number: 11,010 (range: 11,010 to 12,009)
Segment 3 ==> sequence number: 12,010 (range: 12,010 to 13,009)
Segment 4 ==> sequence number: 13,010 (range: 13,010 to 14,009)
Segment 5 ==> sequence number: 14,010 (range: 14,010 to 16,009)

TCP FEATURES

Numbering System:

- Although the TCP software keeps track of the segments being transmitted or received, there is no field for a segment number value in the segment header.
- Instead, there are two fields called the sequence number and the acknowledgment number.
- These two fields refer to the byte number and not the segment number

CONTD…

Byte Number

- TCP numbers all data bytes that are transmitted in a connection.
- Numbering is independent in each direction.
- When TCP receives bytes of data from a process, it stores them in the sending buffer and numbers them.
- The numbering does not necessarily start from O.
- Instead, TCP generates a random number between 0 and 2^{32} - 1 for the number of the first byte.

CONTD…

- For example, if the random number happens to be 1057 and the total data to be sent are 6000 bytes, the bytes are numbered from 1057 to 7056.
- Byte numbering is used for flow and error control.

Sequence Number

- After the bytes have been numbered, TCP assigns a sequence number to each segment that is being sent.
- The sequence number for each segment is the number of the first byte carried in that segment.



Note:

The bytes of data being transferred in each connection are numbered by TCP. The numbering starts with a randomly generated number.



Note:

The value of the sequence number field in a segment defines the number of the first data byte contained in that segment.

CONTD…

Acknowledgment Number

- The communication in TCP is full duplex; when a connection is established, both parties can send and receive data at the same time.
- Each party numbers the bytes, usually with a different starting byte number. The sequence number in each direction shows the number of the first byte carried by the segment.
- Each party also uses an acknowledgment number to confirm the bytes it has received.
- The acknowledgment number defines the number of the next byte that the party expects to receive.

CONTD…

- In addition, the acknowledgment number is cumulative, which means that the party takes the number of the last byte that it has received, safe and sound, adds 1 to it, and announces this sum as the acknowledgment number.
- The term cumulative here means that if a party uses 5643 as an acknowledgment number, it has received all bytes from the beginning up to 5642.



Note:

The value of the acknowledgment field in a segment defines the number of the next byte a party expects to receive. The acknowledgment number is cumulative.

CONTD…

Flow Control

- TCP, unlike UDP, provides flow control. The receiver of the data controls the amount of data that are to be sent by the sender.
- This is done to prevent the receiver from being overwhelmed with data.
- The numbering system allows TCP to use a byte-oriented flow control.

CONTD…

Error Control

- To provide reliable service, TCP implements an error control mechanism.
- Although error control considers a segment as the unit of data for error detection (loss or corrupted segments), error control is byte-oriented.

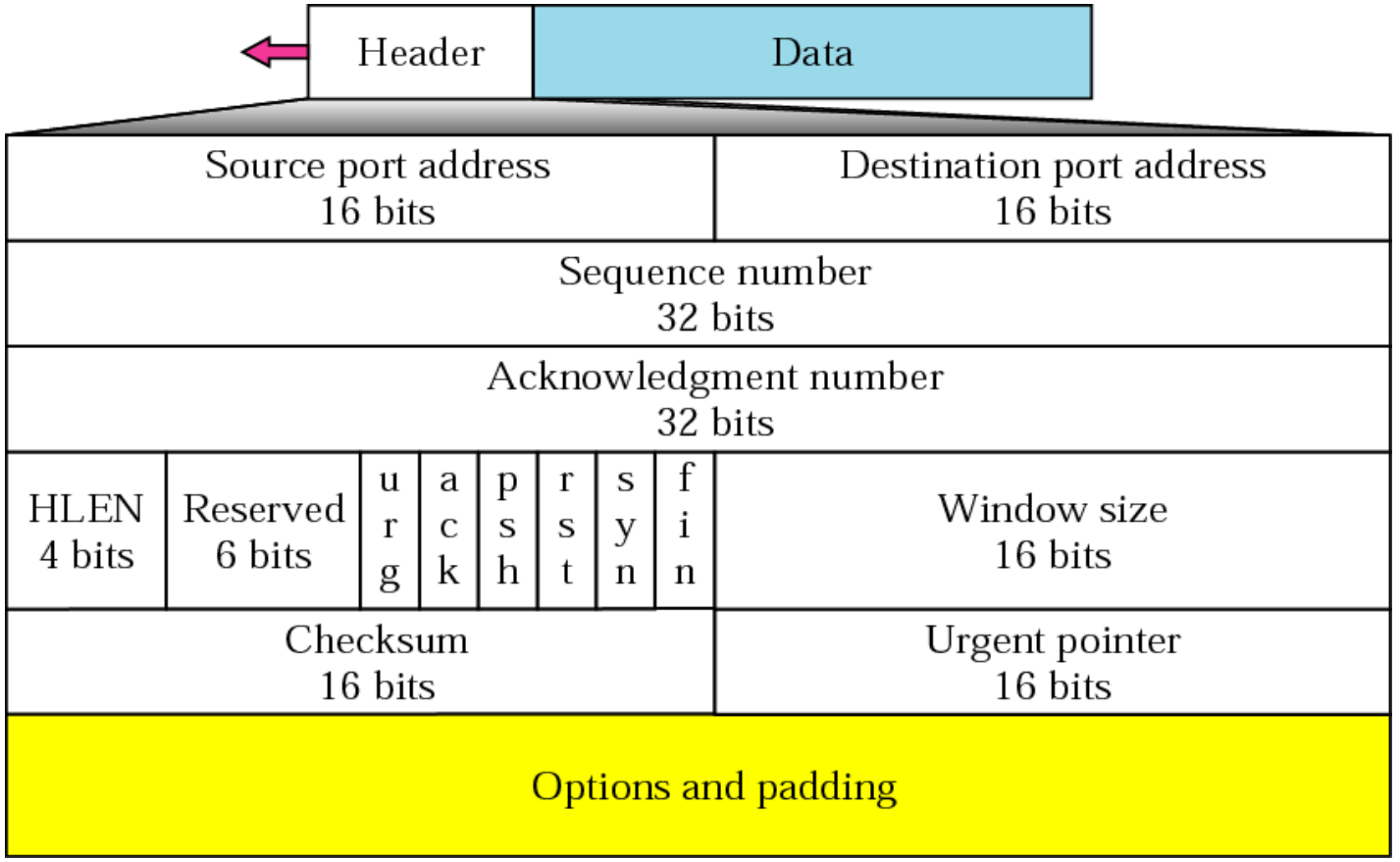
Congestion Control

- TCP, unlike UDP, takes into account congestion in the network.
- The amount of data sent by a sender is not only controlled by the receiver (flow control), but is also determined by the level of congestion in the network.

SEGMENT

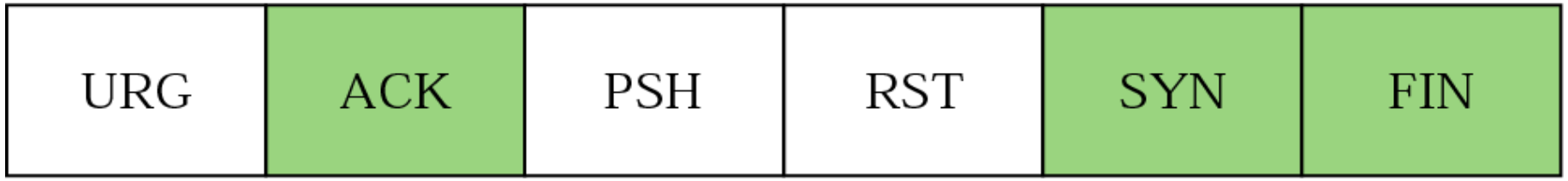
- A packet in TCP is called a segment.
- The segment consists of a 20- to 60-byte header, followed by data from the application program.
- The header is 20 bytes if there are no options and up to 60 bytes if it contains options.

TCP segment format



Control field

URG: Urgent pointer is valid
ACK: Acknowledgment is valid
PSH: Request for push
RST: Reset the connection
SYN: Synchronize sequence numbers
FIN: Terminate the connection



CONTD…

Source port address

- This is a 16-bit field that defines the port number of the application program in the host that is sending the segment. This serves the same purpose as the source port address in the UDP header.

Destination port address

- This is a 16-bit field that defines the port number of the application program in the host that is receiving the segment. This serves the same purpose as the destination port address in the UDP header.

Sequence number

- This 32-bit field defines the number assigned to the first byte of data contained in this segment.

CONTD…

- TCP is a stream transport protocol. To ensure connectivity, each byte to be transmitted is numbered.
- The sequence number tells the destination which byte in this sequence comprises the first byte in the segment.
- During connection establishment, each party uses a random number generator to create an initial sequence number (ISN), which is usually different in each direction.

Acknowledgment number

- This 32-bit field defines the byte number that the receiver of the segment is expecting to receive from the other party.

CONTD…

- If the receiver of the segment has successfully received byte number x from the other party, it defines x + 1 as the acknowledgment number. Acknowledgment and data can be piggybacked together.

Header length

- This 4-bit field indicates the number of 4-byte words in the TCP header. The length of the header can be between 20 and 60 bytes. Therefore, the value of this field can be between 5 (5 x 4 =20) and 15 (15 x 4 =60).

Reserved

- This is a 6-bit field reserved for future use.

Control

- This field defines 6 different control bits or flags as shown in Figure. One or more of these bits can be set at a time

Description of flags in the control field

Flag	Description
URG	The value of the urgent pointer field is valid.
ACK	The value of the acknowledgment field is valid.
PSH	Push the data.
RST	The connection must be reset.
SYN	Synchronize sequence numbers during connection.
FIN	Terminate the connection.

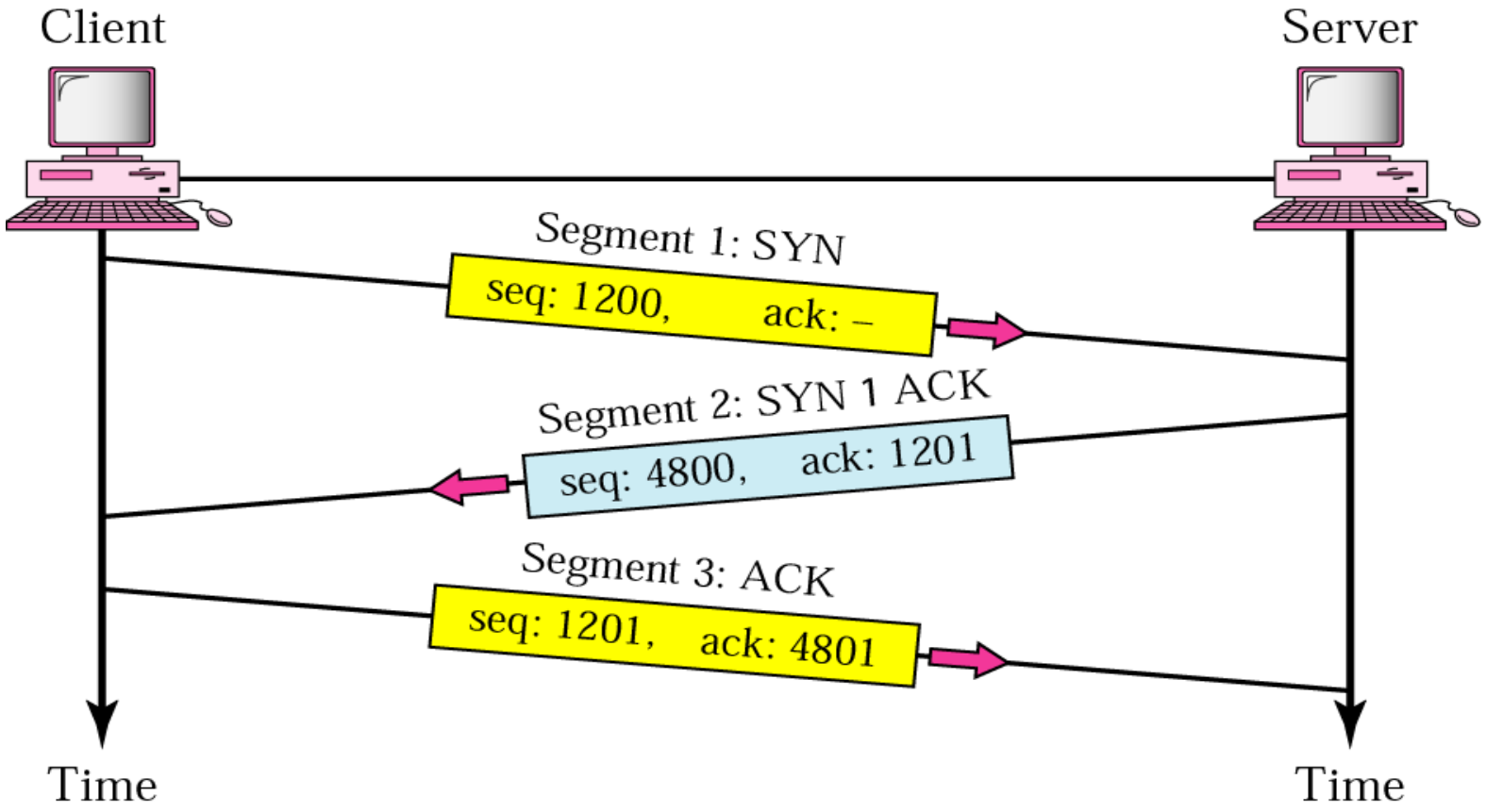
TCP CONNECTION

- A connection-oriented transport protocol establishes a virtual path between the source and destination. All the segments belonging to a message are then sent over this virtual path.
- Using a single virtual pathway for the entire message facilitates the acknowledgment process as well as retransmission of damaged or lost frames.
- How TCP, which uses the services of IP, a connectionless protocol, can be connection-oriented. The point is that a TCP connection is virtual, not physical.
- TCP operates at a higher level. TCP uses the services of IP to deliver individual segments to the receiver, but it controls the connection itself.

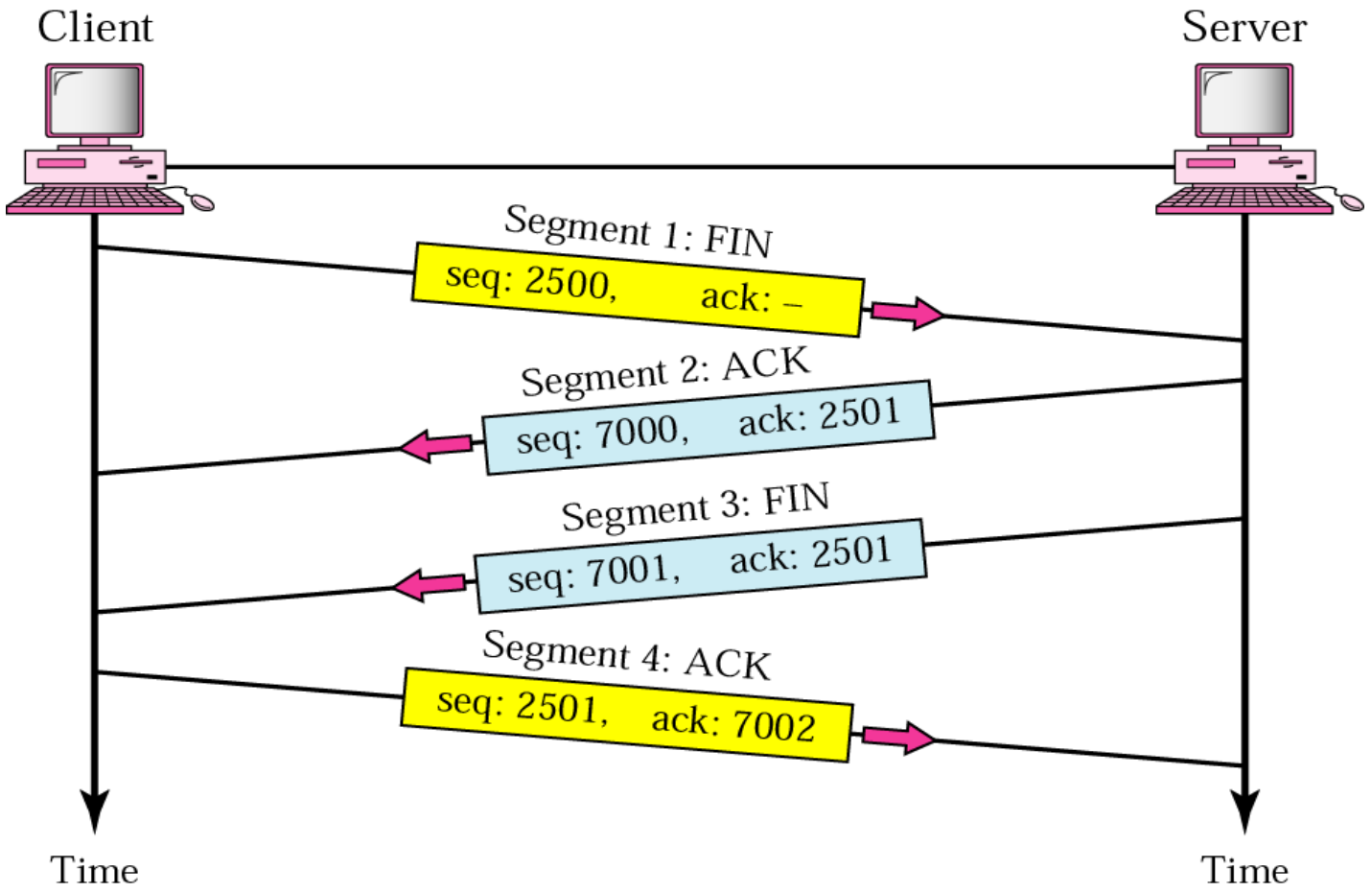
CONTD…

- If a segment is lost or corrupted, it is retransmitted.
- Unlike TCP, IP is unaware of this retransmission.
- If a segment arrives out of order, TCP holds it until the missing segments arrive; IP is unaware of this reordering.
- In TCP, connection-oriented transmission requires three phases: connection establishment, data transfer, and connection termination

Three-step connection establishment



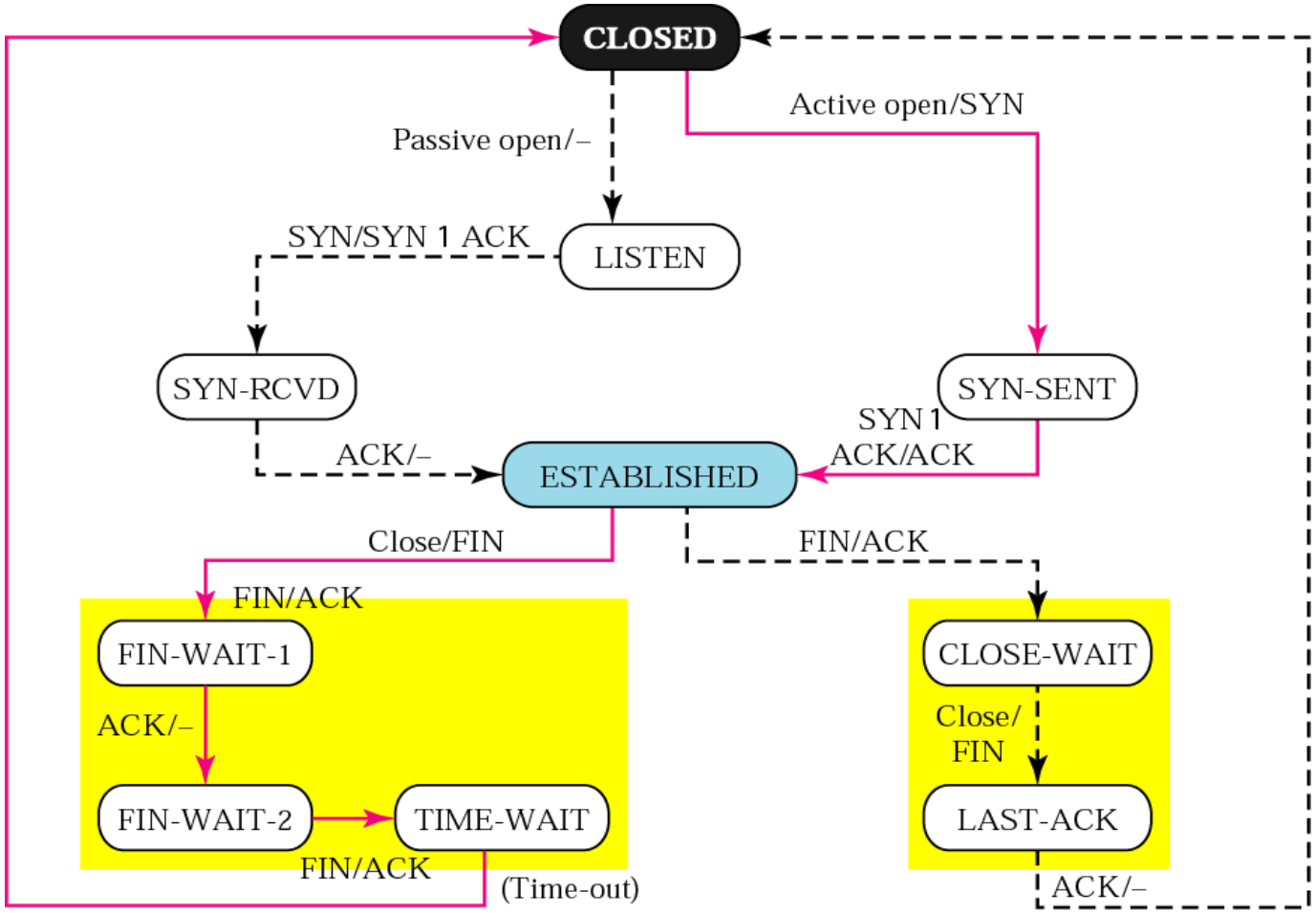
Four-step connection termination



States for TCP

State	Description
CLOSED	There is no connection.
LISTEN	The server is waiting for calls from the client.
SYN-SENT	A connection request is sent; waiting for acknowledgment.
SYN-RCVD	A connection request is received.
ESTABLISHED	Connection is established.
FIN-WAIT-1	The application has requested the closing of the connection.
FIN-WAIT-2	The other side has accepted the closing of the connection.
TIME-WAIT	Waiting for retransmitted segments to die.
CLOSE-WAIT	The server is waiting for the application to close.
LAST-ACK	The server is waiting for the last acknowledgment.

State transition diagram

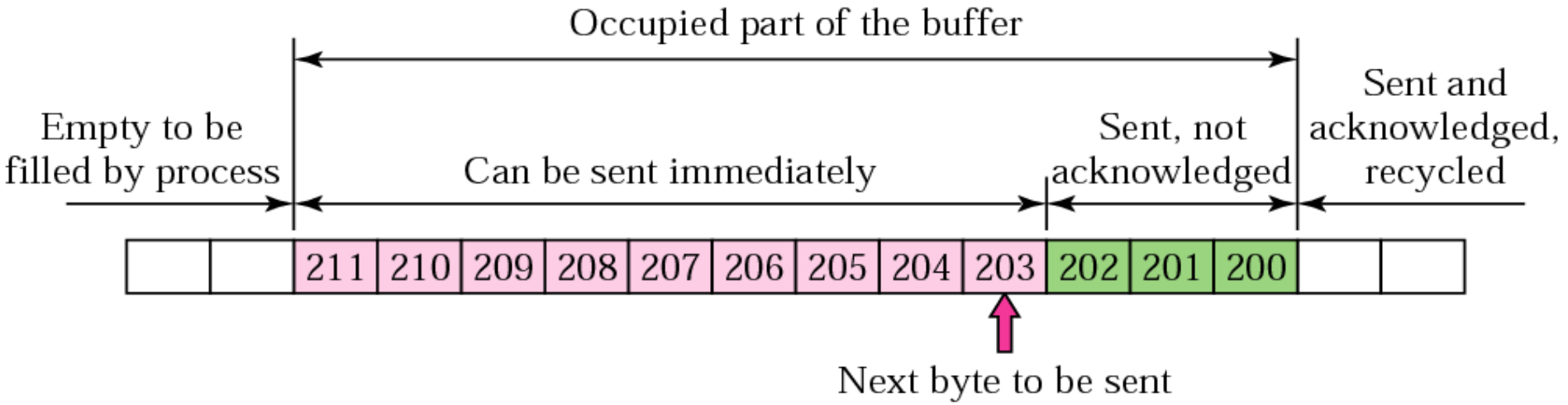




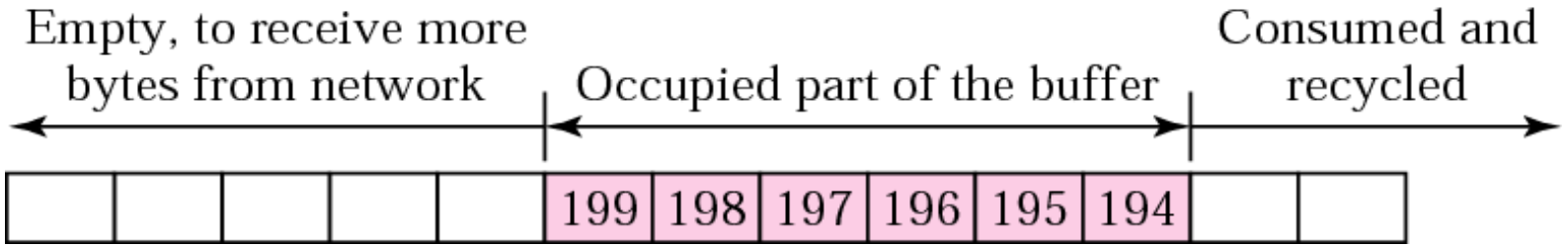
Note:

A sliding window is used to make transmission more efficient as well as to control the flow of data so that the destination does not become overwhelmed with data. TCP's sliding windows are byte-oriented.

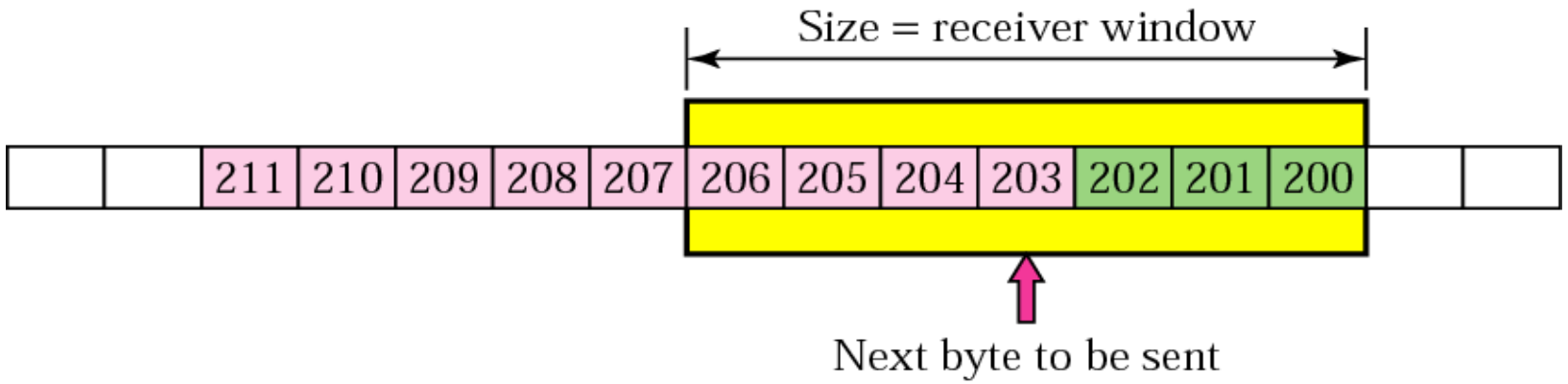
Sender buffer



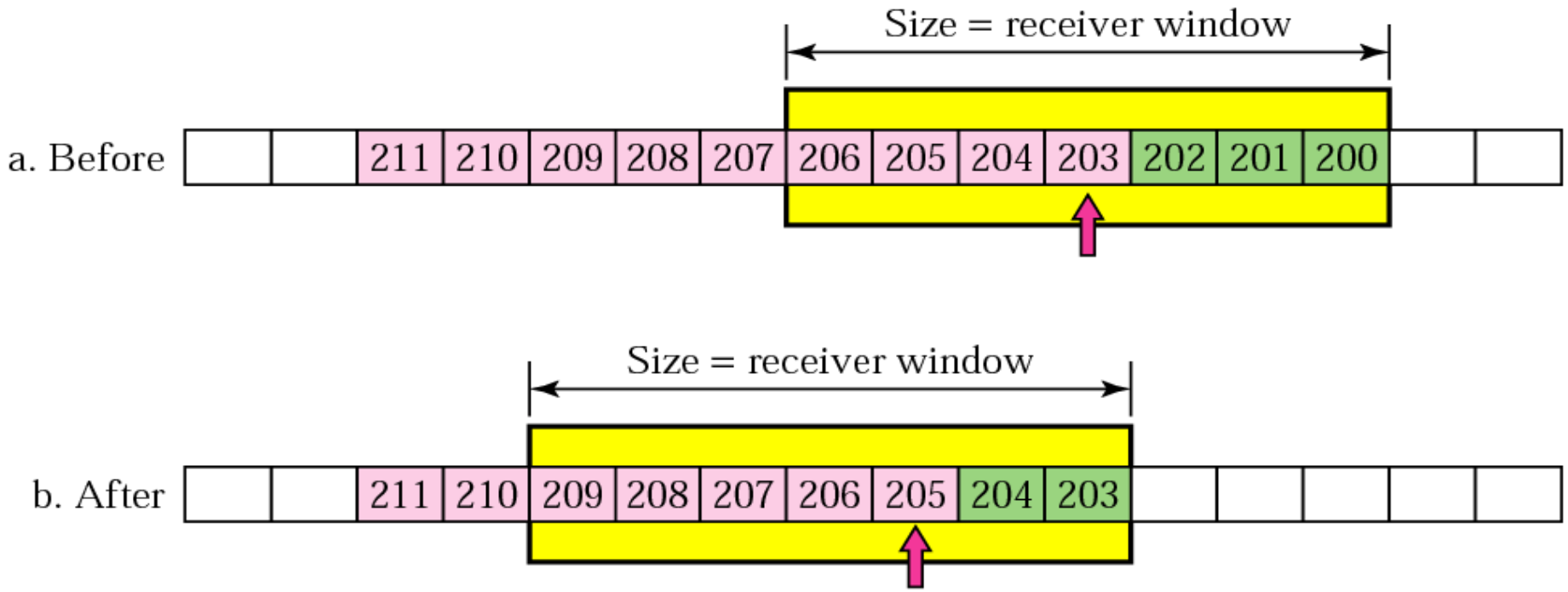
Receiver window



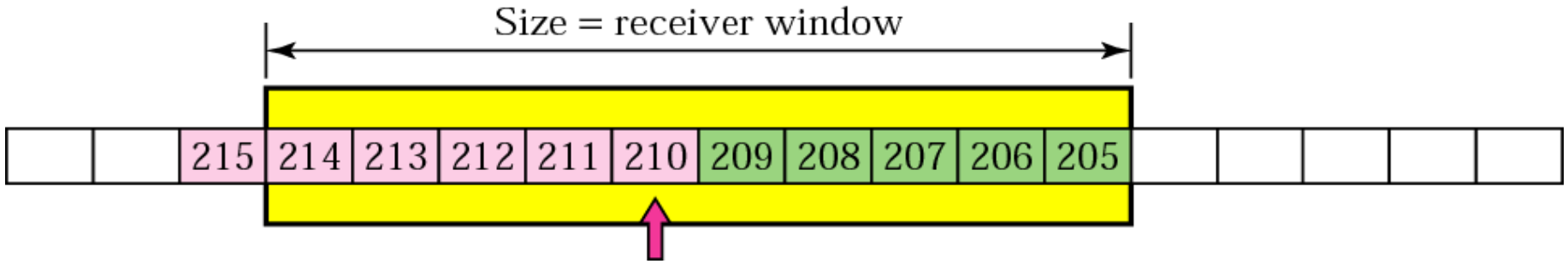
Sender buffer and sender window



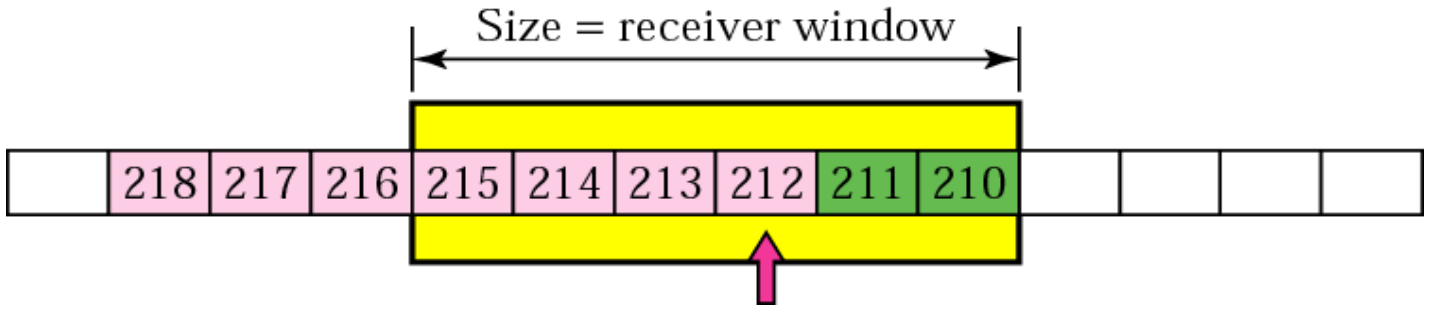
Sliding the sender window



Expanding the sender window



Shrinking the sender window





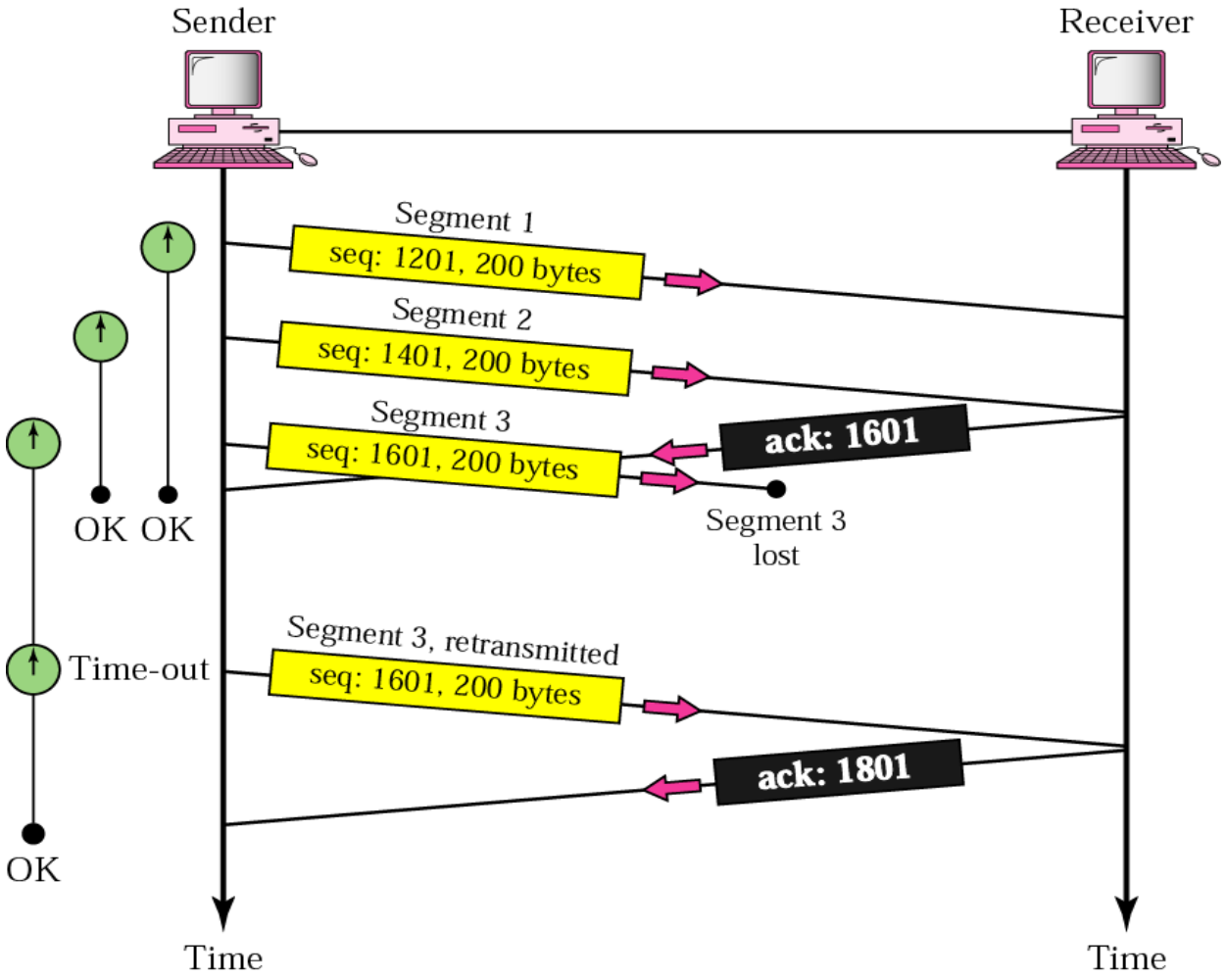
In TCP, the sender window size is totally controlled by the receiver window value (the number of empty locations in the receiver buffer). However, the actual window size can be smaller if there is congestion in the network.



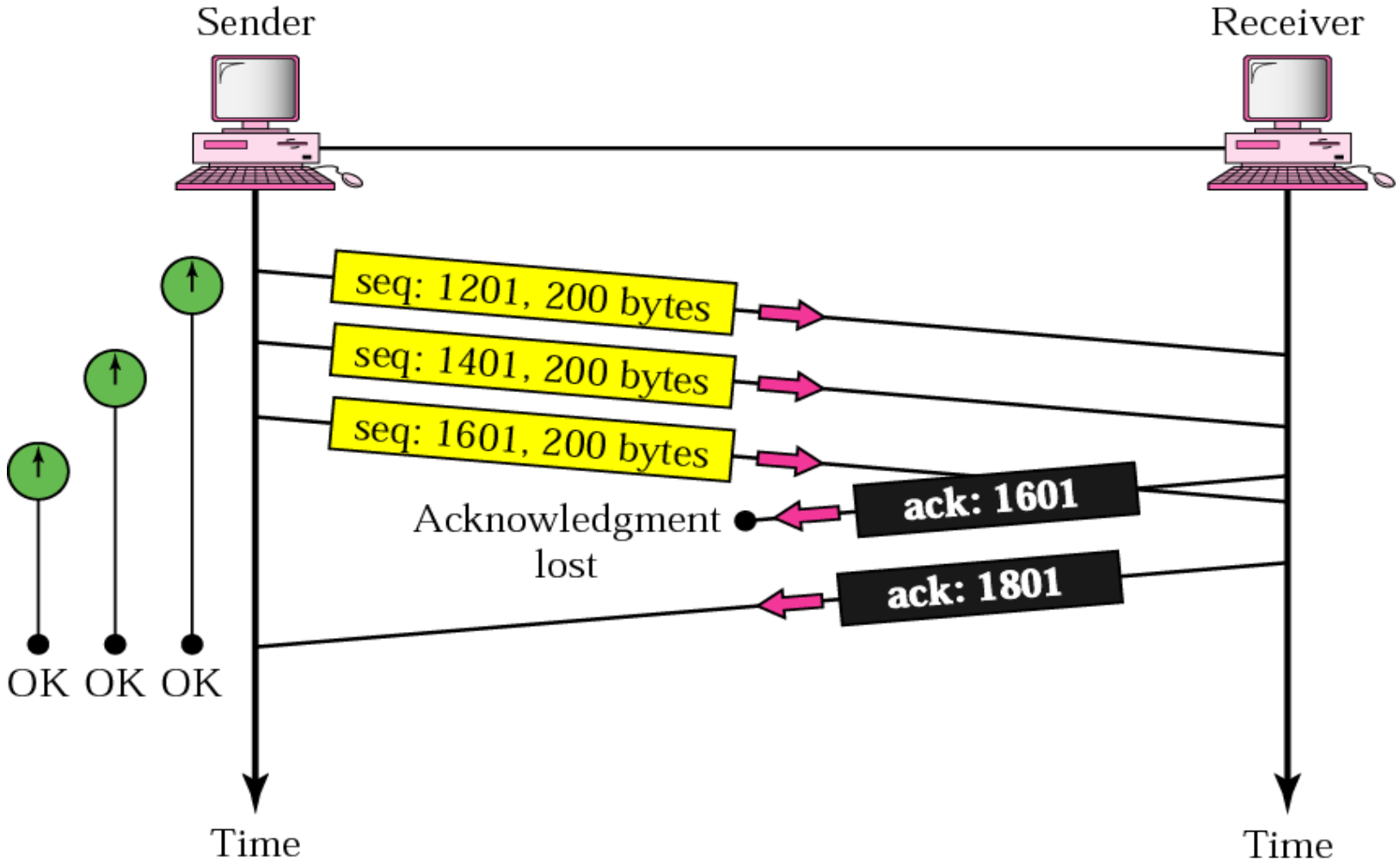
Some points about TCP's sliding windows:

- **The source does not have to send a full window's worth of data.**
- **The size of the window can be increased or decreased by the destination.**

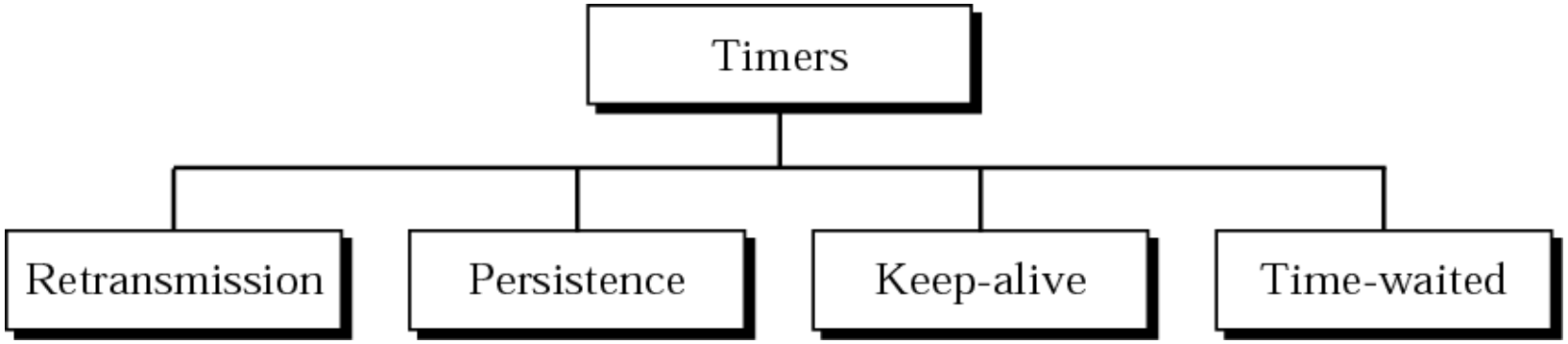
Lost segment



Lost acknowledgment

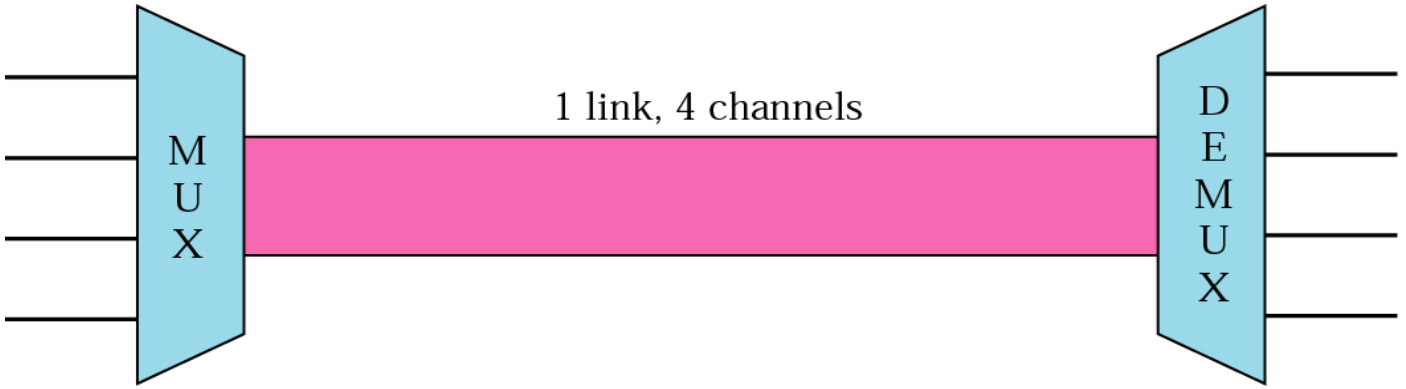


TCP timers


Timers
Retransmission
Persistence
Keep-alive
Time-waited

Multiplexing

Dividing a link into channels



Categories of multiplexing

